

Field Content:

ψ_m^α : Majorana gravitino

Massless: $p^\mu \rightarrow k^\mu = (E, 0, \dots, E)$

irreps of little group $SO(D-2)$

$\rightarrow$ Gravitino multiplet $(\psi_m^\alpha, g_{\mu\nu})$

Has ψ_m^α
 $(D-3) \cdot 2^{[(D-2)/2]}$

$g_{\mu\nu}$
 $\frac{(D-2)^2 - (D-2)}{2} - 1 \approx (D-2)$

$\circ$ grav: $D(D-3)/2$

$\Rightarrow$ (D.O.F)

Fermionic

$(D-3) \cdot 2^{[(D-2)/2]}$

Bosonic:

$\frac{D(D-3)}{2}$

D=11:

$8 \cdot 2^4$

128

$11 \cdot 4$

44

$\rightarrow$ Missing 84 Bosonic D.O.F.

* Can describe Bosons as p-form gauge fields

$A_{\mu_1 \dots \mu_p}$, rank(p) < D transform under pth of $SO(D-2)$

D.O.F. $\binom{D-2}{p}$

D=11: $\binom{9}{p} = \frac{9!}{p! (9-p)!}$

$\Rightarrow$ p=3: $\frac{9!}{3! 6!} = \frac{9 \cdot 8 \cdot 7}{3} = 3 \cdot 4 \cdot 7 = 12 \cdot 7 = 84$

Full theory:

$\begin{pmatrix} g_{\mu\nu} \\ A_{\mu\nu\rho} \\ \psi_m^\alpha \end{pmatrix} \begin{matrix} [2, 0, 0, 0]_q \\ [0, 0, 1, 0]_q \\ [1, 0, 0, 2]_q \end{matrix}$

on-shell



• 3-form Potential $A_{\mu\nu\rho}$ must be invariant under gauge transl. w gauge parameter $\Theta_{\mu\nu\rho}$

$\Theta_{\mu\nu\rho}$: 2-form

* In general, the C.S. terms are of the form

$$S_p = -\frac{1}{2} \int * F^{(p+1)} \wedge F^{(p+1)} \quad | \quad F^{(p+1)} \equiv \delta A^p$$

$$= -\frac{1}{2} \frac{1}{(p+1)!} \int d^D x \sqrt{-g} F^{\mu_1 \dots \mu_{p+1}} F_{\mu_1 \dots \mu_{p+1}}$$

and so we have $p=3$:

$$S_3 = -\frac{1}{2} \frac{1}{4!} \int d^D x \sqrt{-g} F^{\mu_1 \dots \mu_4} F_{\mu_1 \dots \mu_4}$$

$$= \left(-\frac{1}{2}\right) \left(\frac{1}{24}\right) \int d^D x \sqrt{-g} F^{\mu_1 \dots \mu_4} F_{\mu_1 \dots \mu_4}$$

$$S_3 = \frac{1}{2} \int d^D x \sqrt{-g} \left(-\frac{1}{24} F^{\mu_1 \dots \mu_4} F_{\mu_1 \dots \mu_4} \right)$$

$$\delta A_{\mu\nu\rho} = 3 \partial_{[\mu} \Theta_{\nu\rho]}$$

$$F_{\mu\nu\rho\sigma} = 4 \partial_{[\mu} A_{\nu\rho\sigma]}$$

$$\partial_{[\mu} F_{\nu\rho\sigma]} = 0 \quad (\text{Bianchi})$$

$$\uparrow \quad dF = 0 \quad \text{since} \quad d(\delta A) = \delta^2 A = 0$$

* Spinor spin flip:

$$\bar{\chi} \Gamma^A \lambda$$

$$\bar{\chi} \gamma^{\mu_1 \dots \mu_r} \lambda = \text{tr} \bar{\chi} \gamma^{\mu_1 \dots \mu_r} \chi$$

$$t_0 = t_3 = I$$

$$t_1 = t_2 = -I$$

$$\text{tr} t_4 = \text{tr}$$

Base action:

$$S = \frac{1}{2\kappa} \int d^D x \, e \left[e_a^\mu e_b^\nu R_{\mu\nu}{}^{ab} - \bar{\Psi}_\mu \gamma^{\mu\nu\rho} D_\nu \Psi_\rho - \frac{1}{24} F^{\mu_1 \dots \mu_4} F_{\mu_1 \dots \mu_4} + \dots \right]$$

* Must find $+\dots$ terms to leave the action invariant under local susy.

→ Work in second order formalism: $\omega_\mu{}^{ab}(e)$ (torsionless)

• Similar to $D=4$, can find action via Noether method + dim analysis:

$$\delta e_m^\alpha = \frac{1}{2} \bar{\epsilon} \gamma^\alpha \psi_m$$

$$\delta \psi_m = D_m \epsilon$$

these are what was found for GR + gravitino and must be altered / incorporate

• Note: in the second order formalism

$$\delta e_{\mu\nu} = \frac{1}{2} \bar{\epsilon} (\gamma_\mu \psi_\nu + \gamma_\nu \psi_\mu) = \bar{\epsilon} \gamma_{(\mu} \psi_{\nu)}$$

and so, we expect, when looking at $\delta A_{\mu\nu\rho}$ to have terms $\sim \gamma_{[\mu\nu} \psi_{\rho]}$

• Indeed

$$\delta A_{\mu\nu\rho} = -c \bar{\epsilon} \gamma_{[\mu\nu} \psi_{\rho]}$$

• Will determine c : look at flat interacting (A, ψ) theory. Will also see:

$$\delta \psi_\mu = D_\mu \epsilon + (a \gamma^{\alpha\beta\gamma\delta}{}_\mu + b \gamma^{\beta\gamma\delta} \delta_\mu^\alpha) F_{\alpha\beta\gamma\delta} \epsilon$$

$$S_0 = \frac{1}{2} \int d^4x \left[-\bar{\psi}_\mu \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho - \frac{1}{24} F^{\mu\nu\rho\sigma} F_{\mu\nu\rho\sigma} \right]$$

→ Also with global susy

$$\chi_{\mu_1 \dots \mu_5} = \gamma_{\mu_1 \dots \mu_5} \lambda \rightarrow \bar{\chi}_{\mu_1 \dots \mu_5} = \bar{\lambda} \gamma_{\mu_1 \dots \mu_5} \text{totr}$$

$$\therefore \delta \bar{\psi}_\mu = (a \bar{\epsilon} \text{tot}_5 \gamma^{\alpha\beta\gamma\delta}{}_\mu + b \bar{\epsilon} \text{tot}_3 \gamma^{\beta\gamma\delta} \delta_\mu^\alpha) F_{\alpha\beta\gamma\delta}$$

$$\delta \bar{\psi}_\mu = \bar{\epsilon} (-a \gamma^{\alpha\beta\gamma\delta}{}_\mu + b \gamma^{\beta\gamma\delta} \delta_\mu^\alpha) F_{\alpha\beta\gamma\delta}$$

$$\delta \psi_\mu = (a \gamma^{\alpha\beta\gamma\delta}{}_\mu + b \gamma^{\beta\gamma\delta} \delta_\mu^\alpha) \epsilon F_{\alpha\beta\gamma\delta}$$

$$\delta A_{\mu\nu\rho} = -c \bar{\epsilon} \gamma_{[\mu\nu} \psi_{\rho]}$$

$$\begin{aligned} \delta F_{\mu\nu\rho\sigma} &= 4 \partial_{[\mu} \delta A_{\nu\rho\sigma]} \\ &= -4c \bar{\epsilon} \partial_{[\mu} \gamma_{\nu\rho} \psi_{\sigma]} \end{aligned}$$

$$\delta S_0 = \frac{1}{2} \int d^4x \left(-\delta \bar{\psi}_\mu \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho - \bar{\psi}_\mu \gamma^{\mu\nu\rho} \partial_\nu \delta \psi_\rho - \frac{1}{24} F^{\mu\nu\rho\sigma} \delta F_{\mu\nu\rho\sigma} \right)$$

$$\bar{\psi}_\mu \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho = \partial_\nu \bar{\psi}_\rho \gamma^{\mu\nu\rho} \psi_\mu \quad t_3$$

$$\mu\nu\rho = \rho\mu\nu = -\rho\nu\mu$$

$$\Rightarrow \int -\partial_\nu \bar{\psi}_\mu \gamma^{\mu\nu\rho} \psi_\rho d^4x = \int d^4x \partial_\mu (\quad) + \int d^4x (\quad)$$

$$= \int d^4x \delta \bar{\psi}_\mu \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho$$

• Get an extra factor of 2

$$\delta S_0 = \int d^4x \left(-\delta \bar{\psi}_\mu \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho - \frac{1}{24} F^{\mu\nu\rho\sigma} \delta F_{\mu\nu\rho\sigma} \right)$$

$$F^{\mu\nu\rho\sigma} (\partial_\mu \delta A_{\nu\rho\sigma} - \partial_\nu \delta A_{\rho\sigma\mu} + \partial_\rho \delta A_{\sigma\mu\nu} - \partial_\sigma \delta A_{\mu\nu\rho})$$

$$= 4 F^{\mu\nu\rho\sigma} \partial_\mu \delta A_{\nu\rho\sigma}$$

$$= -4c \bar{E} F^{\mu\nu\rho\sigma} \partial_\mu (\gamma_{\nu\rho} \psi_\sigma)$$

$$\therefore \int d^4x \left(-\frac{1}{24} F^{\mu\nu\rho\sigma} \delta F_{\mu\nu\rho\sigma} \right) = \int d^4x \left(-\frac{1}{24} F^{\mu\nu\rho\sigma} \right) (-4c \bar{E} \partial_\mu \gamma_{\nu\rho} \psi_\sigma)$$

$$= \frac{1}{6} \bar{E} c \gamma_{\nu\rho} \int d^4x F^{\mu\nu\rho\sigma} \partial_\mu \psi_\sigma$$

$$= \int d^4x -\frac{1}{6} \bar{E} c \gamma_{\nu\rho} \psi_\sigma \partial_\mu F^{\mu\nu\rho\sigma}$$

$$\therefore \delta S_0 = \bar{E} \int d^4x \left[\left(a \gamma^{\alpha\beta\gamma\delta}{}_\mu - b \gamma^{\alpha\beta\gamma\delta} \partial_\mu \right) F_{\alpha\beta\gamma\delta} \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho \right. \\ \left. - \frac{1}{6} c \gamma_{\nu\rho} \psi_\sigma \partial_\mu F^{\mu\nu\rho\sigma} \right]$$

*

$$\gamma^{\mu_1 \dots \mu_n \rho} \gamma_{\rho \nu_1 \nu_2} = (D-6) \gamma^{\mu_1 \dots \mu_n}{}_{\nu_1 \nu_2} + 8(D-5) \gamma^{\mu_1 \dots \mu_n}{}_{[\nu_2} \delta_{\nu_1]}^{\mu_2} \\ + 12(D-4) \gamma^{[\mu_1 \mu_2} \delta^{\mu_3}{}_{\nu_1} \delta^{\mu_4}{}_{\nu_2]}$$

$$\gamma^{\alpha\beta\gamma\delta}{}_\mu \gamma^{\mu\nu\rho} F_{\alpha\beta\gamma\delta} = (D-6) \gamma^{\alpha\beta\gamma\delta\nu\rho} F_{\alpha\beta\gamma\delta} + 8(D-5) \gamma^{\alpha\beta\gamma}{}_{[\nu} F^{\rho]}{}_{\alpha\beta\gamma} \\ - 12(D-4) \gamma^{\alpha\beta} F_{\alpha\beta}{}^{\nu\rho}$$

and

$$\gamma^{\alpha\beta\gamma\delta} \gamma^{\mu\nu\rho} F_{\alpha\beta\gamma\delta} = -\gamma^{\nu\rho\alpha\beta\gamma\delta} F_{\alpha\beta\gamma\delta} - 6 \gamma^{\alpha\beta\gamma}{}_{[\nu} F^{\rho]}{}_{\alpha\beta\gamma} + 6 \gamma^{\alpha\beta} F_{\alpha\beta}{}^{\nu\rho}$$

1st:

$$\begin{aligned}
 & a \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho (D-6) \gamma^{\alpha\beta\gamma\delta\mu\rho} F_{\alpha\beta\gamma\delta} \\
 & + a \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho 8(D-5) \gamma^{\alpha\beta\gamma[\mu} F^{\rho]}_{\alpha\beta\gamma} \\
 & - a 12(D-4) \gamma^{\mu\nu\rho} \partial_\nu \psi_\rho \gamma^{\alpha\beta} F_{\alpha\beta}{}^{\mu\rho}
 \end{aligned}$$

Rank 6

$$\begin{aligned}
 & -a (D-6) \gamma^{\alpha\beta\gamma\delta\mu\rho} \partial_\nu F_{\alpha\beta\gamma\delta} \psi_\rho - b \gamma^{\nu\rho\alpha\beta\gamma\delta} \partial_\nu F_{\alpha\beta\gamma\delta} \psi_\rho \\
 & = 5! \partial_\nu F_{\alpha\beta\gamma\delta} = 0 \quad (\text{Bianchi})
 \end{aligned}$$

Rank 4

$$-a 8(D-5) \gamma^{\alpha\beta\gamma[\mu} \partial_\nu F^{\rho]}_{\alpha\beta\gamma} \psi_\rho - b 6 \gamma^{\alpha\beta\gamma[\mu} \partial_\nu F^{\rho]}_{\alpha\beta\gamma}$$

$$\Rightarrow 8(D-5)a + 6b = 0$$

Rank 2 $12a(D-4) \gamma^{\alpha\beta} \partial_\nu F_{\alpha\beta}{}^{\mu\rho} \psi_\rho + 6b \gamma^{\alpha\beta} \partial_\nu F_{\alpha\beta}{}^{\mu\rho}$

$$\Rightarrow \gamma_{\nu\alpha} \psi_\rho \partial_\mu F^{\nu\sigma\mu\rho} = \gamma^{\alpha\beta} \psi_\rho \partial_\nu F_{\alpha\beta}{}^{\mu\rho}$$

$$(12a(D-4) + 6b - \frac{1}{6}c) \gamma^{\alpha\beta} \psi_\rho \partial_\nu F_{\alpha\beta}{}^{\mu\rho} = 0$$

$$12(D-4)a + 6b = \frac{1}{6}c$$

D=11: $8a + b = 0 \Rightarrow$

$$b = -8a$$

$$12 \cdot 7a + 6b = \frac{1}{6}c$$

$$a = c/216$$

*

$$\begin{aligned}
 \delta \psi_\mu &= \partial_\mu \epsilon + \frac{c}{216} (\gamma^{\alpha\beta\gamma\delta} \partial_\mu - 8 \gamma^{\beta\gamma\delta} \partial_\mu^\alpha) F_{\alpha\beta\gamma\delta} \epsilon \\
 \delta A_{\mu\nu\rho} &= -c \epsilon \gamma_{\mu\nu\rho} \psi_\rho
 \end{aligned}$$

* $F_{\mu\nu\rho\sigma} = 0$

* Fix ϵ by closure of $\{\omega, \omega\} = F$

$$[\delta_1, \delta_2] A_{mnp} = \delta_1 \delta_2 A_{mnp}$$

$$\begin{aligned} & \delta_1 (-c \bar{E}_1 \gamma_{[mn} \psi_{p]}) - (1 \leftrightarrow 2) \\ &= -c \bar{E}_1 \gamma_{[mn} \delta_1 \psi_{p]} - [] \\ & \quad c \bar{E}_2 \gamma_{mn} \left(\frac{c}{216} [\gamma^{\alpha\beta\gamma\delta} \delta_{\rho}] - 8 \gamma^{\beta\gamma\delta} \delta_{\rho}^{\alpha} \right) F_{\alpha\beta\gamma\delta} E_1 \\ &= -\frac{c^2}{216} \bar{E}_2 \gamma_{[mn} (\gamma^{\alpha\beta\gamma\delta} \delta_{\rho}] - \gamma^{\beta\gamma\delta} \delta_{\rho}^{\alpha}) F_{\alpha\beta\gamma\delta} E_1 \end{aligned}$$

$$(i) \bar{E}_2 () E_1 - \bar{E}_2 () E_2$$

but

$$\bar{E}_2 () E_2 = \text{tr} \bar{E}_1 () E_2$$

$\leadsto$ only rank 1 orb 5 since $t_1 = t_5 = -1$
 (ii) $\bullet 1^{\text{st}}$ term has no rank 1 part from $[]$

$$(iii) \gamma^{mnp} \gamma_{\sigma\tau} = \gamma^{mnp}_{\sigma\tau} + 6 \gamma^{[mn} \delta_{[\sigma} \delta_{\tau]}^{\rho]} + 6 \gamma^{[m} \delta_{[\sigma}^{\nu} \delta_{\tau]}^{\rho]} \quad \text{only contribution}$$

$$\begin{aligned} &= -\frac{c^2}{108} \bar{E}_2 \gamma_{[mn} \gamma^{\beta\gamma\delta} \delta_{\rho]} F_{\alpha\beta\gamma\delta} E_2 \\ &= -\frac{c^2}{108} \bar{E}_1 6 \gamma^{[B} \delta_{[L}^{\gamma} \delta_{M}^{\delta]} \delta_{\rho]}^{\alpha} F_{\alpha\beta\gamma\delta} \\ &= -\frac{c^2}{18} 2 \cdot 4 \bar{E}_1 \gamma^B F_{\rho\beta\gamma\delta} \end{aligned}$$

$$\therefore = -\frac{4c^2}{9} \bar{E}_1 \gamma^{\sigma} E_2 F_{\sigma mnp}$$

$$[\delta_1, \delta_2] A_{mnp} = -\frac{4c^2}{9} \bar{E}_1 \gamma^{\sigma} E_2 F_{\sigma mnp}$$

• Rank 5 terms cancel

Note: $\{\delta_{E_1}, \delta_{E_2}\} \sim \delta_5^{\text{S.C.}}$

$$F_{\sigma mnp} = \partial_{\sigma} A_{mnp} + (-\partial_m A_{np\sigma} + \partial_n A_{p\sigma m} - \partial_p A_{\sigma mn})$$

$-c^2 = \gamma^{\sigma\tau} \gamma_{\sigma\tau}$

$$\Rightarrow \frac{1}{4} \bar{E}_1 \gamma^\sigma E_2 \partial_\sigma A_{\mu\nu\rho} = \underbrace{\left(-\frac{4c^2}{9} \bar{E}_2 \gamma^\sigma E_2 \right) \partial_\sigma A_{\mu\nu\rho}}_{\text{Space-time translation}} + \underbrace{\frac{-4c^2}{9} \bar{E}_1 \gamma^\sigma E_2 \left(\right)}_{\text{Gauge-transf}}$$

$$\xi^\sigma \equiv -\frac{1}{2} \bar{E}_1 \gamma^\sigma E_2 \quad \Bigg| \quad \Theta_{\mu\nu} \equiv -\bar{E}_1 \gamma^\sigma E_2 A_{\sigma\mu\nu}$$

$$[\delta_1, \delta_2] A_{\mu\nu\rho} = -\frac{c^2}{9} \xi^\sigma \partial_\sigma A_{\mu\nu\rho} + \frac{c^2}{9} (\partial_\mu \Theta_{\nu\rho} + \partial_\nu \Theta_{\rho\mu} + \partial_\rho \Theta_{\mu\nu})$$

$$\frac{1}{2} = \frac{4c^2}{9} \Rightarrow \boxed{c^2 = \frac{9}{8}} \Rightarrow c = \frac{3}{2\sqrt{2}}$$

$$\delta \psi_\mu = \partial_\mu \epsilon + \frac{\sqrt{2}}{288} (\gamma^{\alpha\beta\gamma\delta}{}_\mu - 8 \gamma^{\rho\gamma\delta} \delta^\alpha_\mu) F_{\alpha\beta\gamma\delta} \epsilon$$

$$\delta A_{\mu\nu\rho} = -\frac{\sqrt{2}}{4} \bar{\epsilon} \gamma_{[\mu\nu} \psi_{\rho]}$$

* Find Super-current: Local

$$\delta S_0 = \int d^4x \left(-\partial_\mu \bar{\epsilon} \right) \left(\right) - \frac{c}{6} \gamma_{\mu\rho} \psi_\rho F^{\mu\rho\sigma\tau}$$

$$a = \frac{\sqrt{2}}{288}, \quad c = \frac{3\sqrt{2}}{4}, \quad b = -\frac{\sqrt{2}}{36}$$

$$\delta S_0 = \int d^4x \partial_\mu \bar{\epsilon} \left[\left(-\frac{\sqrt{2}}{288} \dots \right) \right]$$

Claim: Rank 4 terms cancel

$$= \int d^4x \partial_\mu \bar{\epsilon} \left[-\frac{\sqrt{2}}{288} \left(5 \gamma^{\alpha\beta\gamma\delta\sigma\mu\rho} F_{\alpha\beta\gamma\delta} - 12 \cdot 7 \gamma^{\alpha\beta} F_{\alpha\beta}^{\mu\rho} \right) \psi_\rho \right. \\ \left. - \frac{\sqrt{2}}{36} \left(-\gamma^{\mu\rho\alpha\beta\gamma\delta} F_{\alpha\beta\gamma\delta} + 6 \gamma^{\alpha\beta} F_{\alpha\beta}^{\mu\rho} \right) \psi_\rho \right. \\ \left. + \frac{\sqrt{2} \cdot 3}{4 \cdot 6} \gamma_{\mu\sigma} \psi_\rho F^{\sigma\mu\rho\tau} \right]$$

Rank 6

$$F_{\alpha\beta\gamma\delta} \left(-\frac{5\sqrt{2}}{288} \gamma^{\alpha\beta\gamma\delta\sigma\mu\rho} + \frac{\sqrt{2}}{288} \gamma^{\mu\rho\alpha\beta\gamma\delta} \right) \psi_\rho$$

$$\alpha_{\beta\gamma\delta\mu\rho} = \mu\alpha_{\beta\gamma\delta\rho} = \mu\rho\alpha_{\beta\gamma\delta\rho}$$

$$\begin{aligned} &= F_{\alpha\beta\gamma\delta} \gamma^{\mu\rho\alpha\beta\gamma\delta} \psi_{\rho} \frac{\sqrt{2}}{36} \left(\frac{-5}{8} + 1 \right) \\ &= \left(\frac{\sqrt{2}}{36} \frac{3}{8} \right) () = \boxed{\frac{\sqrt{2}}{96} F_{\alpha\beta\gamma\delta} \gamma^{\alpha\beta\gamma\delta\mu\rho} \psi_{\rho}} \end{aligned}$$

Rank 2]

$$\left(\frac{\sqrt{2} 12.7}{8.36} - \frac{\sqrt{2} 6}{36} \right) \gamma^{\alpha\beta} F_{\alpha\beta}{}^{\mu\rho} \psi_{\rho} + \frac{\sqrt{2} 3}{4.6} \gamma^{\alpha\beta} F_{\alpha\beta}{}^{\mu\rho} \psi_{\rho}$$

wrong factor somewhere....

$$\boxed{\frac{\sqrt{2}}{8} \gamma^{\alpha\beta} F_{\alpha\beta}{}^{\mu\rho} \psi_{\rho}}$$

$$= \int d^4x \partial_{\mu} \bar{E} \frac{\sqrt{2}}{96} (F_{\alpha\beta\gamma\delta} \gamma^{\alpha\beta\gamma\delta\mu\rho} + 12 \gamma^{\alpha\beta} F_{\alpha\beta}{}^{\mu\rho}) \psi_{\rho}$$

$$\therefore \mathcal{T}^{\mu} = \frac{\sqrt{2}}{96} (\gamma^{\alpha\beta\gamma\delta\mu\rho} F_{\alpha\beta\gamma\delta} + 12 \gamma^{\alpha\beta} F_{\alpha\beta}{}^{\mu\rho}) \psi_{\rho}$$

* Super current of the susy $\partial_{\mu} \bar{E} \rightarrow -\bar{\Psi}_{\mu}$

$$\int d^4x \partial_{\mu} \bar{E} \mathcal{T}^{\mu} \Rightarrow S_1 = S_0 - \frac{1}{2k^2} \int d^4x \bar{\Psi}_{\mu} \mathcal{T}^{\mu}$$

$$\therefore S_1 = \frac{1}{2k^2} \int d^4x e \left[e^{a\mu} e^{b\nu} R_{\mu\nu\alpha\beta} - \bar{\Psi}_{\mu} \gamma^{\mu\nu\rho} D_{\nu} \psi_{\rho} - \frac{1}{24} F^{\mu_1 \dots \mu_5} F_{\mu_1 \dots \mu_5} \right. \\ \left. - \frac{\sqrt{2}}{96} \bar{\Psi}_{\mu} (\gamma^{\alpha\beta\gamma\delta\mu\rho} F_{\alpha\beta\gamma\delta} + 12 \gamma^{\alpha\beta} F_{\alpha\beta}{}^{\mu\rho}) \psi_{\rho} + \dots \right]$$

with

$$\begin{aligned} \delta e_{\mu}^a &= \frac{1}{2} \bar{E} \gamma^a \psi_{\mu} \\ \delta \psi_{\mu} &= D_{\mu} \bar{E} + \frac{\sqrt{2}}{288} (\gamma^{\alpha\beta\gamma\delta}{}_{\mu} - 8 \gamma^{\beta\gamma\delta} \delta_{\mu}^{\alpha}) F_{\alpha\beta\gamma\delta} \bar{E} \\ \delta A_{\mu\nu\rho} &= -\frac{3\sqrt{2}}{4} \bar{E} \gamma_{\mu\nu} \psi_{\rho} \end{aligned}$$

$$\delta A_{\mu\nu\rho} \sim \bar{E} R_{\mu\nu}{}^{\rho\sigma} \gamma^{\sigma} \psi_{\rho}$$

and $\sim \bar{E} F_{\alpha\beta\gamma\delta} \psi$ cancel

• Curvature terms cancel.

• $D_\nu \bar{E} \gamma^\nu$ is cancelled by $\delta \bar{\psi}_\nu = D_\nu \bar{E}$ term

$$S = \frac{1}{k^2} \int d^4x e L$$

→ Consider $e F^2 \psi$ terms:

$$\delta L_{FF} : \delta e F^2$$

$$\delta L_{\bar{\psi} F \psi} : \delta \psi \sim F e \text{ in } \delta S_0$$

$$e = \det(e^a_\mu) = e^{\text{tr} h(e)}$$

$$\begin{aligned} \delta e &= \text{tr}(e^{-1} \delta e) e \\ &= e e^\mu_a \delta e^\mu_a \end{aligned}$$

$$\delta L_{FF} = \frac{1}{2} \int d^4x e e^\sigma_a \left(\frac{1}{2} \bar{E} \gamma^\sigma \psi \right) \left(-\frac{1}{24} \right) F^{\mu\nu\rho\sigma} F_{\mu\nu\rho\sigma}$$

$$+ \frac{1}{2} \int d^4x e \left(-\frac{1}{24} \right) 8 \delta e^\mu_m F^{\mu\nu\rho\sigma} F_{\mu\nu\rho\sigma}$$

$$= \frac{1}{2} \int d^4x e \left[\left(-\frac{1}{24} \right) \left(\frac{1}{2} \bar{E} \gamma^\sigma \psi \right) 8 F^{\mu\nu\rho\sigma} F_{\mu\nu\rho\sigma} + e^\sigma_a \left(\frac{1}{2} \bar{E} \gamma^\sigma \psi \right) F^2 \right]$$

$$\textcircled{1} = \frac{1}{2} \int d^4x e \left(-\frac{1}{48} \right) \left(\bar{E} \gamma^\sigma \psi \right) F^{\mu\nu\rho\sigma} F_{\mu\nu\rho\sigma}$$

$$= \frac{1}{48} \int d^4x e \left(-\frac{1}{2} \right) g^{\mu\nu} \bar{E} \gamma_\sigma \psi F^{\rho\sigma\tau} F_{\nu\rho\sigma\tau}$$

$$\textcircled{2} = \frac{1}{48} \int d^4x 8 \delta e^\mu_m F^{\mu\rho\sigma\tau} F_{\rho\sigma\tau}$$

$$+ \frac{1}{48} \int d^4x 8 \frac{1}{2} \bar{E} \gamma_a \psi^\mu F^{\alpha\rho\sigma\tau} F_{\mu\rho\sigma\tau}$$

$$= \frac{1}{48} \int d^4x 4 \bar{E} \gamma^\mu \psi^\nu F_{\alpha\rho\sigma\tau} F_{\nu\rho\sigma\tau}$$

$$= \frac{1}{48} \int d^4x 4 \bar{E} e^\mu_m \gamma^\mu \psi^\nu F_{\alpha\rho\sigma\tau} F_{\nu\rho\sigma\tau}$$

$$= \frac{1}{48} \int d^4x 4 \bar{E} \gamma^\mu \psi^\nu F_{\mu\rho\sigma\tau} F_{\nu\rho\sigma\tau}$$

$$\therefore \delta L_{FF} \sim \frac{1}{48} \left(4 \bar{E} \gamma^\mu \psi^\nu - \frac{1}{2} g^{\mu\nu} \bar{E} (\gamma \cdot \psi) \right) F_{\mu\rho\sigma\tau} F_{\nu\rho\sigma\tau}$$

$$\frac{1}{48} \bar{E} (\gamma^{\alpha'\beta'\gamma'\delta'} + 8 \gamma^{\beta'\delta'\delta'\zeta\alpha'}) F_{\alpha'\beta'\gamma'\delta'} F_{\zeta\alpha'\beta'\gamma'}$$

$$\delta L_{FF\psi} = 96 \times 144 \epsilon \left(\gamma^{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta} F_{\alpha\beta\gamma\delta} + 12 \gamma^{\alpha\beta} F_{\alpha\beta}{}^{\gamma\delta} \right) \psi_\rho$$

Get rank 9, 7, 5, 3, 1 terms

1, 3, 5, 7 terms cancel (most)

→ Look only at rank 9 terms:

$$\gamma^{\alpha'\beta'\gamma'\delta'} \gamma^{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta} = (D-9) \gamma^{\alpha'\beta'\gamma'\delta'\alpha\beta\gamma\delta\epsilon} + \dots$$

$$\gamma^{\beta'\gamma'\delta'} \gamma^{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta} = \frac{2}{\gamma^{\alpha'\beta'\gamma'\delta'\alpha\beta\gamma\delta\epsilon}} + \dots$$

$$\text{Rank 9: } (\delta L_{FF} + \delta L_{FF\psi}) = -\frac{1}{16 \times 144} \gamma^{\alpha'\beta'\gamma'\delta'\alpha\beta\gamma\delta\epsilon} \psi_\rho F_{\alpha'\beta'\gamma'\delta'} F_{\alpha\beta\gamma\delta\epsilon}$$

For odd dim ($D=11$): 32×32 -matrices

$$\gamma^0, \dots, \gamma^{10} = \gamma^*, \quad \gamma_* = \gamma^0 \gamma^1 \dots \gamma^9 \quad (5=m)$$

then

$$\gamma^{\mu_1 \dots \mu_r} = i^{m+1} \frac{1}{(D-r)!} \epsilon^{\mu_1 \dots \mu_r \mu_{r+1} \dots \mu_D} \gamma_{\mu_{r+1} \dots \mu_D}$$

$$\gamma^{\alpha'\beta'\gamma'\delta'\alpha\beta\gamma\delta\epsilon} = -\frac{1}{2\epsilon} \epsilon^{\alpha'\beta'\gamma'\delta'\alpha\beta\gamma\delta\epsilon\mu\nu} \gamma_{\mu\nu}$$

$$e(\delta L_{FF} + \delta L_{FF\psi}) = -\frac{1}{2} \epsilon^{\alpha'\beta'\gamma'\delta'\alpha\beta\gamma\delta\epsilon\mu\nu} \gamma_{\mu\nu} \left(\frac{1}{16 \times 144} \right) \psi_\rho F_{\alpha'\beta'\gamma'\delta'} F_{\alpha\beta\gamma\delta\epsilon}$$

$$e(\delta L_{FF} + \delta L_{FF\psi}) = \frac{1}{32 \times 144} (\epsilon^{\alpha'\alpha}) (-\bar{\epsilon} \gamma_{\mu\nu} \psi_\rho F_{\alpha'\beta'\gamma'\delta'} F_{\alpha\beta\gamma\delta\epsilon})$$

$$\delta A_{\mu\nu\rho} = -\frac{3\sqrt{2}}{4} \bar{\epsilon} \gamma_{\mu\nu} \psi_\rho$$

$$e(\delta L_{FF} + \delta L_{FF\psi}) = \left(\frac{4}{3\sqrt{2}} \right) \left(\frac{1}{32 \times 144} \right) \epsilon^{\alpha'\alpha} (\delta A_{\mu\nu\rho}) F_{\alpha'\beta'\gamma'\delta'} F_{\alpha\beta\gamma\delta\epsilon}$$

∴ We can add (minus) this term to the action

$$S_{CS} = -\frac{\sqrt{2}}{(144 \times 144)^2} \int d^4 x \epsilon^{\alpha'\beta'\gamma'\delta'\alpha\beta\gamma\delta\epsilon\mu\nu\rho} F_{\alpha'\beta'\gamma'\delta'} F_{\alpha\beta\gamma\delta\epsilon} A_{\mu\nu\rho}$$

$$= -\frac{\sqrt{2}}{6K^2} \int F^{(4)} \wedge F^{(4)} \wedge A^{(3)}$$

$$\therefore F^{(4)} = F \wedge F$$

(Chern-Simons)

$$= \frac{\sqrt{2}}{6k^2} \int dA^{(2)} \wedge dA^{(2)} \wedge A^{(3)}$$

$$\begin{aligned} \delta \left(\int F \wedge F \wedge A \right) &= \int d\delta A \wedge F \wedge A + \int F \wedge d(\delta A) \wedge A \\ &\quad + \int F \wedge F \wedge \delta A \\ &= \int \delta A \wedge F \wedge F + \int F \wedge \delta A \wedge dA + \int F \wedge F \wedge \delta A \\ &= \int 2\delta A \wedge F \wedge F + \int F \wedge F \wedge \delta A \\ &= 3 \int F \wedge F \wedge \delta A \end{aligned}$$

* Does not produce further variations, since δ frame fields

* Also invariant under gauge transformations

$$\delta A^{(3)} = d\theta^{(2)}$$

$$\begin{aligned} \delta S_{\text{CS}} &= 3 \int F^{(4)} \wedge F^{(4)} \wedge d\theta^{(2)} \\ &= 0 \quad (\text{from } dF^{(4)} = 0 \text{ [Bianchi]}) \end{aligned}$$

Final Action:

$$\begin{aligned} S = \frac{1}{2\kappa^2} \int d^4x \left[\left(e^{a\mu} e^{b\nu} R_{\mu\nu\alpha\beta} - \bar{\Psi}_\mu \gamma^{\mu\nu\rho} D_\nu \left(\frac{1}{2}(\omega + \hat{\omega}) \right) \Psi_\rho - \frac{1}{24} F^{\mu\nu\rho\sigma} F_{\mu\nu\rho\sigma} \right. \right. \\ \left. \left. - \frac{\sqrt{2}}{192} \bar{\Psi}_\nu (\gamma^{\alpha\beta\gamma\delta} + 12 \gamma^{\alpha\beta} g^{\gamma\delta}) \Psi_\rho (F_{\alpha\beta\gamma\delta} + \hat{F}_{\alpha\beta\gamma\delta}) \right. \right. \\ \left. \left. - \frac{2\sqrt{2}}{(4\kappa)^2} \epsilon^{\alpha'\beta'\gamma'\delta'\alpha\beta\gamma\delta\mu\nu\rho} F_{\alpha'\beta'\gamma'\delta'} F_{\alpha\beta\gamma\delta} A_{\mu\nu\rho} \right] \end{aligned}$$

$$\omega_{\mu ab} = \omega_{\mu ab}(e) + K_{\mu ab}$$

$$\hat{\omega}_{\mu ab} = \omega_{\mu ab}(e) - \frac{1}{4} (\bar{\Psi}_\mu \gamma_b \Psi_a - \bar{\Psi}_a \gamma_\mu \Psi_b + \bar{\Psi}_b \gamma_a \Psi_\mu)$$

$$K_{\mu ab} = -\frac{1}{4} (\bar{\Psi}_\mu \gamma_b \Psi_a - \bar{\Psi}_a \gamma_\mu \Psi_b + \bar{\Psi}_b \gamma_a \Psi_\mu) + \frac{1}{8} \bar{\Psi}_\nu \gamma^{\nu\rho} \omega_{\mu ab} \Psi_\rho$$

$$\hat{F}_{\mu\nu\rho\sigma} = 4 \partial_{[\mu} A_{\nu\rho\sigma]} + \frac{3}{2} \sqrt{2} \bar{\Psi}_{[\mu} \gamma_{\nu\rho} \Psi_{\sigma]}$$

$$\delta e_\mu^a = \frac{1}{2} \bar{\epsilon} \gamma^a \psi_\mu$$

$$\delta \psi_\mu = D_\mu(\hat{\omega}) \epsilon + \frac{\sqrt{2}}{288} (\gamma^{\alpha\beta\gamma\delta}{}_\mu - 8 \gamma^{\beta\gamma\delta} \delta_\mu^\alpha) \hat{F}_{\alpha\beta\gamma\delta} \epsilon$$

$$\delta A_{\mu\nu\rho} = -\frac{3\sqrt{2}}{4} \bar{\epsilon} \gamma_{[\mu\nu} \psi_{\rho]}$$

* Flat connections are supercovariant: quantity whose local SUSY does not contain the derivative of $\partial_\mu \epsilon$

→ E.O.M. must transform super consistently

E.O.M of gravitino:

$$\gamma^{\mu\nu\rho} D_\nu \psi_\rho - \frac{\sqrt{2}}{288} \gamma^{\mu\nu\rho} (\gamma^{AB\sigma\delta} \psi_\sigma - 8 \gamma^{B\sigma\delta} \psi_\sigma^A) \psi_\rho \hat{F}_{AB\sigma\delta} = 0$$

$$* [\delta_Q(E_1), \delta_Q(E_2)] = \delta_{\text{gct}}(\xi^\mu) + \delta_L(\lambda^{ab}) + \delta_Q(E_3) + \delta_A(b_{uv})$$

$\uparrow$ Lorentz $\nearrow$ susy $\nwarrow$ 3-form gauge

$$\begin{aligned} \xi^\mu &\equiv \frac{1}{2} \bar{E}_2 \gamma^\mu E_2 \\ \lambda^{ab} &\equiv -\xi^\mu \hat{\omega}_\mu^{ab} + \frac{\sqrt{2}}{288} \bar{E}_1 (\gamma^{ab\mu\nu\rho\sigma} F_{\mu\nu\rho\sigma} + 24 \gamma_{\mu\nu} \hat{F}^{ab\mu\nu}) E_2 \\ E_3 &\equiv -\xi^\mu \psi_\mu \\ \Theta_{uv} &\equiv -\xi^\rho A_{\rho uv} + \frac{\sqrt{2}}{4} \bar{E}_1 \gamma_{uv} E_2 \end{aligned}$$

• Satisfied on ψ_μ if it satisfies its E.O.M. constraint
 → Cannot find Auxiliary fields for off-shell

$\bar{E}_1 \Gamma^{(2)} E_2, \bar{E}_2 \Gamma^{(8)} E_2$: do not vanish for classical
 (M2 and M5-brane solutions)
 → Central Charges

$$* 2^{\text{MA}} = 2^S = \underline{32}, \quad \epsilon = 1, \quad B^T = B, \quad C = -C^T$$

Central Charges:

→ Existence of central charges $\leftrightarrow$ (susy preserving) branes

D: Even: $\{Q_\alpha, Q_\beta\} = (\gamma_m C^{-1})_{\alpha\beta} P^m + \sum_{p \geq 1} (\gamma_{m_1 \dots m_p} C^{-1})_{\alpha\beta} Z^{m_1 \dots m_p}$

• $\sum$: over all possible central terms

• Only occur if $(\gamma_{m_1 \dots m_p} C^{-1})_{\alpha\beta}$ are symmetric

$$\alpha\beta = (\alpha\beta)$$

→ Note: $\gamma_m^T = -C \gamma_m C^{-1}$

$$\Rightarrow (C \Gamma^p)^T = -\epsilon_r C \Gamma^{p^T}$$

$$\text{or } \gamma_{m_1 \dots m_p} C^{-1} = \epsilon (-1)^{(p-1)(p-2)/2} (\gamma_{m_1 \dots m_p} C^{-1})^T$$

and so we require

$$\epsilon = (-1)^{(p-1)(p-2)/2} = 1$$

* $\epsilon = 1 \quad (D = 2, 4 \bmod 8), \text{ i.e. } D = 10, \epsilon = 1$
 $(-1)^{(p-1)(p-2)/2} = 1$

• $D = 2, 4, 10$: Majorana spinors

$$\Rightarrow p = 1, 2, 5, 6, 9, 10$$

* $(D = 6, 8 \bmod 8) \quad \epsilon = -1$

$$p = 3, 4, 7, 8, 11, 12$$

Spinors are Weyl: $(P_{\pm} Q)_{\alpha} = 0$

then

$$\{Q_{\alpha}, Q_{\beta}\} = (P_{\pm} \gamma_m C^{-1})_{\alpha\beta} P^m + \sum_p (P_{\pm} \gamma_{m_1 \dots m_p} C^{-1})_{\alpha\beta} Z^{m_1 \dots m_p}$$

• $(\gamma_{m_1 \dots m_p} C^{-1})_{\alpha\beta}$ and $(\gamma^{p+1} \gamma_{m_1 \dots m_p} C^{-1})_{\alpha\beta}$ must be symmetric

$$\leadsto (\gamma^{p+1} \gamma_{m_1 \dots m_p} C^{-1})_{\alpha\beta} = \frac{1}{(p+1)!} (-i)^{m+1} \epsilon_{m_1 \dots m_p n_1 \dots n_{D-p}} \gamma^{n_1 \dots n_{D-p}}$$

• $p = 1, 2 \bmod 4$

$$(D-p) = 1, 2 \bmod 4 \quad \epsilon = 1$$

Majorana-Weyl: $D = 2 \bmod 8$

$$D = 2 + 8n, \quad n \in \mathbb{Z}$$

$$p = 1, 5, 9, 13 \dots$$

ex: $N=1, D=10$ (Super Y.M.) $\mathcal{E}_I$ Type I

$$\{Q_{\alpha}, Q_{\beta}\} = (P_{\pm} \gamma_m C^{-1})_{\alpha\beta} P^m + (P_{\pm} \gamma_{m_1 \dots m_5} C^{-1})_{\alpha\beta} Z^{m_1 \dots m_5} + (P_{\pm} \gamma_{m_1 \dots m_9} C^{-1})_{\alpha\beta} Z^{m_1 \dots m_9}$$

$$\leadsto \epsilon^{m_1 \dots m_9} \gamma_{m_1 \dots m_9} \propto \gamma^9 \gamma_{11}$$

absorb $\epsilon_{m_1 \dots m_9} Z^{m_1 \dots m_9}$ into P_n

$$\{Q_{\alpha}, Q_{\beta}\} = (P_{\pm} \gamma_m C^{-1})_{\alpha\beta} P^m + (P_{\pm} \gamma_{m_1 \dots m_5} C^{-1})_{\alpha\beta} Z^{m_1 \dots m_5}$$

(Type I string)

$\rightarrow$ Self-dual, # of central charges

LO

$$+ \frac{1}{2} \binom{10}{5}$$

$$(\text{from } \epsilon_{m_1 \dots m_p} z^{m_1 \dots m_p}) \sim p m$$

$$= \underline{136}$$

• Maj-wght: $2^5/2 = 16$ components

$$\frac{16 \times 17}{2} = 17 \cdot 8 = \underline{136}$$

(not sure where the 17 comes from)

Odd dim: Do not have Weyl

• $D = 3, 7, \underline{11}$, γ_{D+1} behaves exactly like the other γ under comp cond; $\hat{e}$ transpose

• $D = 5, 9, 13 \dots$ sep.

11 Dim: Majorana Supercharges:

$$p = 1, 2, 5, 6, 9, 10$$

* we only need to go up to rank 5:

$$\gamma_{m_1 \dots m_p} = \frac{(-1)^{p(p-1)/2} (-1)^{q(q-1)/2}}{q!} \epsilon^{m_1 \dots m_p n_1 \dots n_q} \gamma_{n_1 \dots n_q}$$

$$p+q=11, \text{ so}$$

$$\gamma_{m_1 \dots m_p} = - \frac{(-1)^{p(p+3)/2}}{(11-p)!} \epsilon^{m_1 \dots m_p n_1 \dots n_q} \gamma_{n_1 \dots n_q}$$

$$\gamma_{10} \equiv \gamma_0 \dots \gamma_9$$

ex: $\gamma_{m_1 \dots m_5} = - \frac{(-1)^{6 \cdot 0/2}}{5!} \epsilon^{m_1 \dots m_5 n_1 \dots n_5} \gamma_{n_1 \dots n_5}$

$$(\Gamma^{m_1 \dots m_5} C^{-1})_{\alpha\beta} z_{m_1 \dots m_5}$$

$$= \frac{1}{5!} (\epsilon^{m_1 \dots m_5 n_1 \dots n_5} \gamma_{n_1 \dots n_5} C^{-1})_{\alpha\beta} z_{m_1 \dots m_5}$$

$$\sum_{n_1 \dots n_5} = \frac{1}{5!} \epsilon^{m_1 \dots m_5 n_1 \dots n_5} z_{m_1 \dots m_5}$$

$$= (\gamma_{n_1 \dots n_5} C^{-1})_{\alpha\beta} \sum_{n_1 \dots n_5}$$

$$Z = \tilde{Z} + Z_0$$

etc....

M-theory:

$$\{Q_{\alpha}, Q_{\beta}\} = (\gamma^m C^{-1})_{\alpha\beta} P_m + (I^{mn} C^{-1})_{\alpha\beta} Z_{mn} + (I^{mnpqr} C^{-1})_{\alpha\beta} Z_{mnpqr}$$

LHS: 32×32 matrix $\rightarrow$ 528 off-shell

$$\begin{aligned} \text{RHS: } 11 + \binom{11}{2} + \binom{12}{5} \\ = 11 + \left(\frac{12!}{4!2!} + \frac{11!}{6!5!} \right) \\ = 528 \quad \checkmark \end{aligned}$$

* Look at 10d theory when we can reduce the theory

* Massless reps of $SO(D-2)$

Spin 2: Symmetric, traceless rep of little group
 $h_{ij} = h_{ji}, \quad h^i_i = 0$

* Value of $k_{||}^{-2}$ at the classical level has no physical meaning.

• Weyl-rescaling:

$$\begin{aligned} e_m^\mu &\rightarrow e^{-\alpha} e_m^\mu & A^{(3)} &\rightarrow e^{-3\alpha} A^{(3)} \\ \psi_m &\rightarrow e^{\alpha/2} \psi_m & K_\mu &\rightarrow e^{-\alpha/2} K_\mu \end{aligned}$$

$$R(e^\alpha e_m^\mu) = e^{-2\alpha} \left(R(e_m^\mu) + 2(D-1) D^\mu p_\mu \alpha + (D-1)(D-2) (2\alpha)^2 \right)$$

for constant α : $R(e^\alpha e_m^\mu) = e^{-2\alpha}$

• Not a symmetry since we are rescaling the coupling

* Weyl: Symmetry of E.O.M.

$\Rightarrow$ We can only specify K_{33} wrt a particular metric. (or ... could we not just look at eq. classes of $g^{\mu\nu}$?)

(Dynamical mass $m = p$)

$$h k_{||}^{-2} \int \dots$$

counting: $\int d^4x e$

$$d^2s \rightarrow e^{-2\alpha} d^2s$$

• "absorb" $e^{-\theta\alpha}$ in t : $t \rightarrow e^{-\theta\alpha} t$

$\Rightarrow$ Small K_{12} , or eq. small t is the same as working at small proper distance

Going from $- + + + \rightarrow + - - -$

$$Q \rightarrow -Q, \quad \gamma^a \rightarrow i\gamma^a$$

$$R \rightarrow -R, \quad -i/2 \bar{\Psi}_m \gamma^{\mu\nu\rho} D_\mu \Psi_\rho \rightarrow (-1)^3 \left(\frac{-i}{2}\right) (i)^3 \bar{\Psi}_m \gamma^{\mu\nu\rho} D_\mu \Psi_\rho$$
$$= \frac{i^4}{2} (1) = \frac{1}{2}$$